

Lecture 10: Learning from Demonstration

COMP3018/COMP5018: Human-Robot Interaction

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Introduction

- Today's Topics

- 1- Learning from demonstration

- Session Learning Outcomes

By the end of today's lecture you will be able to:

- 1- Describe the learning approach and its categories

Introduction: Why learn from demonstration?

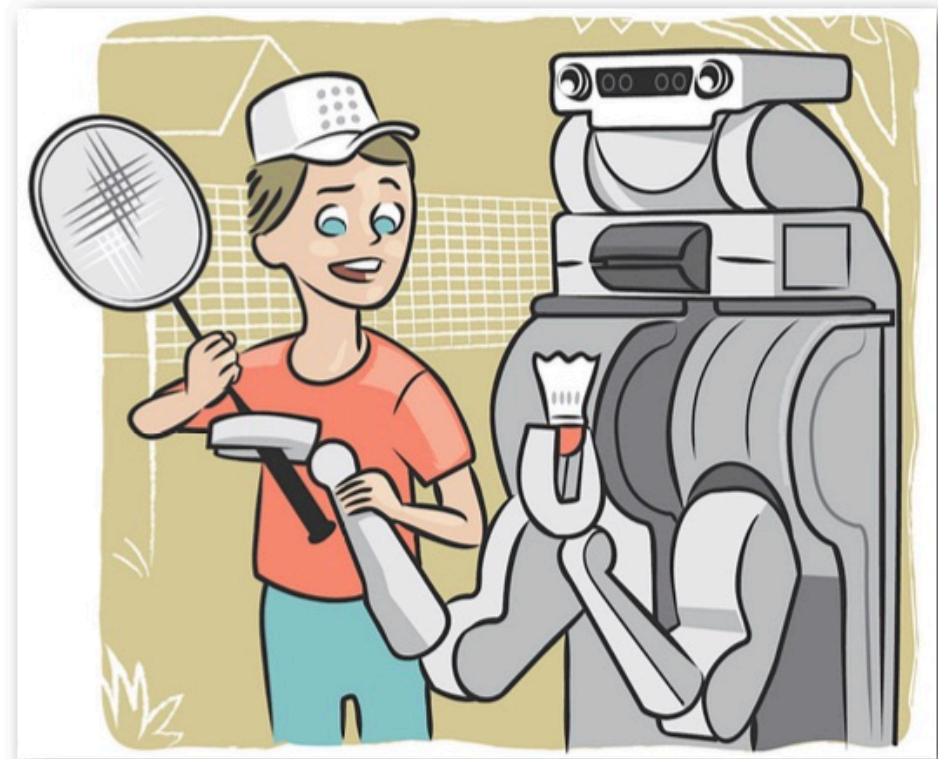
Programming robots is hard!

- Huge number of possible tasks
- Unique environmental demands
- Tasks difficult to describe formally
- Expert engineering impractical



Introduction: Why learn from demonstration?

- Natural, expressive way to program
- No expert knowledge required
- Valuable human intuition
- Program new tasks as-needed



How can robots be shown how to perform tasks?

Learning from Human Demonstrations: Principle

- Transfer to the robot skills that took years for the humans to master.
- Human can quickly re-train the robot to adapt to task changes.
- The human teaches by showing how to perform the task.



Learning from Demonstration (LfD)

- **Learning from Demonstration:** Deriving a policy from examples provided by a teacher. The learner first observes the actions of an (often human) expert, during the training phase. The learner then **uses this training set to learn a policy** that tries to perform the same task demonstrated by the expert, in order to achieve the best performance.
- **It is different from Reinforcement Learning**, in which a policy is derived from experience. This means that **there is no such expert**; the agent has a "**reward**" function, and it uses greedy/exploitative strategies to effectively explore the state and action space, and come up by itself (using trial and error) with an **optimal policy**.



Teacher

demonstration

A screenshot of a data table titled 'syview_1_shoes.sas7bdat'. The table has columns for Region, Product, Subsidiary, Store, Sales, Inventory, and Returns. It contains multiple rows of data, including entries for the United States, Western Europe, and various products like Slippers, Sport Shoes, Women's C, Women's D, Men's Cas, Men's Dre, and Sandals.

	Region	Product	Subsidiary	Store	Sales	Inventory	Returns
322	United States	Slipper	New York	24	\$250,756	\$1,309,20	\$5,623
323	United States	Sport Sho	New York	23	\$22,196	\$141,148	\$847
324	United States	Women's C	New York	10	\$179,842	\$455,655	\$4,255
325	United States	Women's D	New York	17	\$283,972	\$1,125,89	\$16,287
326	United States	Boat	Seattle	15	\$70,793	\$228,629	\$3,898
327	United States	Men's Cas	Seattle	6	\$56,642	\$112,638	\$3,295
328	United States	Men's Dre	Seattle	10	\$132,297	\$274,380	\$5,152
329	United States	Sandal	Seattle	10	\$6,694	\$16,768	\$181
330	United States	Slipper	Seattle	10	\$155,295	\$553,627	\$5,277
331	United States	Sport Sho	Seattle	29	\$89,460	\$169,560	\$1,821
332	United States	Women's C	Seattle	2	\$17,262	\$58,552	\$790
333	United States	Women's D	Seattle	21	\$120,436	\$392,237	\$4,057
334	Western Europe	Boat	Copenhagen	1	\$58,492	\$90,572	\$257
335	Western Europe	Men's Cas	Copenhagen	2	\$12,415	\$26,374	\$684
336	Western Europe	Men's Dre	Copenhagen	4	\$16,080	\$85,133	\$184
337	Western Europe	Slipper	Copenhagen	1	\$1,737	\$17,605	\$17
338	Western Europe	Sport Sho	Copenhagen	15	\$191,927	\$327,742	\$4,294
339	Western Europe	Sport Sho	Copenhagen				

Policy
derivation



Learner

Learning from Demonstration (LfD)

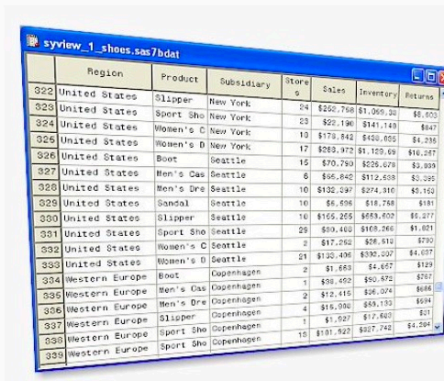
Two phases of LfD:

- Gathering examples: the process of recording example data to derive a policy from
- Deriving policies: analyzing examples to determine a policy



Teacher

demonstration



	Region	Product	Subsidiary	Store	Exes	Inventory	Returns
322	United States	Slipper	New York	24	\$250,750	\$1,069,20	\$0,000
323	United States	Sport Sho	New York	25	\$25,190	\$141,143	\$0,00
324	United States	Women's C	New York	13	\$175,142	\$402,005	\$0,235
325	United States	Women's D	New York	17	\$263,372	\$1,105,19	\$10,207
326	United States	Bout	Seattle	15	\$70,793	\$225,679	\$2,309
327	United States	Men's Cas	Seattle	6	\$66,842	\$112,638	\$2,395
328	United States	Men's Dre	Seattle	10	\$132,297	\$274,310	\$5,162
329	United States	Sandal	Seattle	16	\$8,496	\$18,768	\$181
330	United States	Slipper	Seattle	16	\$165,255	\$553,803	\$6,277
331	United States	Sport Sho	Seattle	29	\$89,403	\$169,566	\$1,091
332	United States	Women's C	Seattle	2	\$17,262	\$56,510	\$790
333	United States	Women's D	Seattle	21	\$120,436	\$330,227	\$4,637
334	Western Europe	Bout	Copenhagen	2	\$1,658	\$4,467	\$129
335	Western Europe	Men's Cas	Copenhagen	1	\$50,302	\$60,572	\$267
336	Western Europe	Men's Dre	Copenhagen	2	\$12,415	\$26,074	\$688
337	Western Europe	Slipper	Copenhagen	4	\$15,090	\$49,193	\$614
338	Western Europe	Sport Sho	Copenhagen	1	\$1,627	\$17,462	\$21
339	Western Europe	Sport Sho	Copenhagen	10	\$181,927	\$627,742	\$4,294

Policy
derivation



Learner

Learning from Demonstration (LfD)

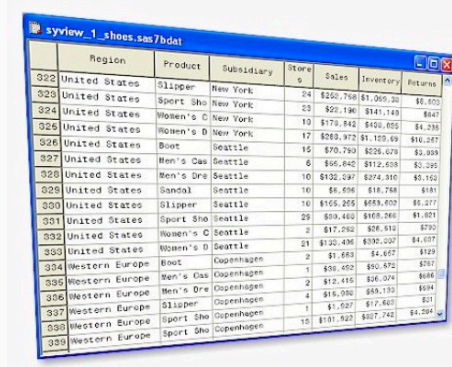
The formal definition of LfD:

- The world consists of states S (it defines the set of possible states of the environment) and actions A (it defines the set of possible actions that an agent can take in the environment).
- States Z are observable states (Z represents the set of demonstrations provided by the human expert) that are mapped from S to Z by mapping M (represents the ultimate objective that the agent is trying to learn).
- The policy $\pi: Z \rightarrow A$ is a selection of actions A based on the observable world states.



Teacher

demonstration

A screenshot of a data table from a software application. The table has columns for Region, Product, Subsidiary, Stores, Sales, Inventory, and Returns. It contains multiple rows of data, including entries for the United States, Western Europe, and Copenhagen.

	Region	Product	Subsidiary	Stores	Sales	Inventory	Returns
322	United States	Slipper	New York	24	\$202,750	\$1,069,30	\$6,670
323	United States	Sport Sho	New York	23	\$22,196	\$141,143	\$647
324	United States	Women's C	New York	10	\$173,842	\$402,802	\$4,285
325	United States	Women's D	New York	17	\$263,372	\$1,125,09	\$16,287
326	United States	Boat	Seattle	15	\$70,793	\$226,679	\$2,898
327	United States	Men's Cas	Seattle	6	\$56,642	\$112,658	\$2,395
328	United States	Men's Dre	Seattle	10	\$132,597	\$274,380	\$5,182
329	United States	Sandal	Seattle	10	\$4,496	\$18,798	\$181
330	United States	Slipper	Seattle	10	\$165,255	\$553,603	\$6,277
331	United States	Sport Sho	Seattle	29	\$69,488	\$168,260	\$1,821
332	United States	Women's C	Seattle	2	\$17,282	\$26,810	\$750
333	United States	Women's D	Seattle	21	\$132,436	\$332,227	\$4,077
334	Western Europe	Boat	Copenhagen	2	\$1,663	\$4,487	\$129
335	Western Europe	Men's Cas	Copenhagen	2	\$12,415	\$16,074	\$684
336	Western Europe	Men's Dre	Copenhagen	4	\$15,090	\$49,133	\$594
337	Western Europe	Slipper	Copenhagen	1	\$1,527	\$17,801	\$21
338	Western Europe	Sport Sho	Copenhagen	15	\$101,927	\$527,742	\$4,294
339	Western Europe	Sport Sho	Copenhagen				

Policy
derivation



Learner

Learning from Demonstration (LfD)

- The policy is derived *after* all training data is obtained (**batch**) or is developed incrementally as data becomes available (**interactive**).
- One of the challenges in LfD is to ensure the **continuity of the problem space**, which means that the agent should be able to generalize the learned behavior to new situations that are similar to the demonstrated ones.
 - In LfD, the **problem space** is defined by the task that the agent needs to learn, the environment in which the task is performed, and the demonstrations provided by the expert.
- **Collect diverse demonstrations:** To ensure that the agent can generalize to different situations, it is essential to **collect a diverse set of demonstrations that cover different variations of the task**. For example, if the task is to navigate a robot in an environment, the demonstrations should cover different types of obstacles, terrains, and lighting conditions.

Learning from Demonstration (LfD)

- Deriving a Policy: Mapping Function
 - It attempts to calculate the underlying function behind the states and actions (input-output pairs provided by a human expert) and generalize over a set of training data. Some common methods used for calculating the underlying function are:
 - ▶ **Regression-based methods:** they involve fitting a function to the input-output pairs using techniques such as linear regression. The resulting function serves as the mapping function.
 - ▶ **Decision tree-based methods:** they involve constructing a tree structure that partitions the input space based on the values of the input features. The output for each partition is determined by the majority vote of the corresponding output values in the training data.
 - Once you have trained the decision tree, you can use it to **estimate the mapping function M** for new input data. Simply pass the input features through the decision tree, and the leaf node reached by the input will correspond to the estimated output value.
 - ▶ **Inverse Reinforcement Learning (IRL)**

Learning from Demonstration (LfD)

- To learn this action policy, a teacher provides a series of examples of the world state and the associated actions
- Often, the world state isn't **directly** accessible to the robot, so it has to depend on **its observations**
- The learned action policy then tells the **robot what action to take based on its observations**
- During learning, knowledge is transferred to the robot
- That transfer may be explicit or implicit
 - **Explicit** if the robot has direct access to the teacher's knowledge
 - the motion and world state information that characterize the actions being taught
 - **Implicit** if the robot has to infer it

Learning from Demonstration (LfD)

There are two steps to this transfer:

1. **Recording** of the teacher's actions
 2. Determining the **correspondence** of this data with the robot's embodiment
- Either step can be direct or indirect

Direct

The data can be used by the robot without modification

Indirect

The data has to be mapped to the robot

i.e. the information required by the robot to reproduce the action
has to be inferred

Learning from Demonstration (LfD)

- Since there are two stages
 - Recording teacher actions
 - Determining correspondence with robot embodiment
- Each with two options
 - Directly used
 - Indirectly mapped
- There are **four different types of learning from demonstration**

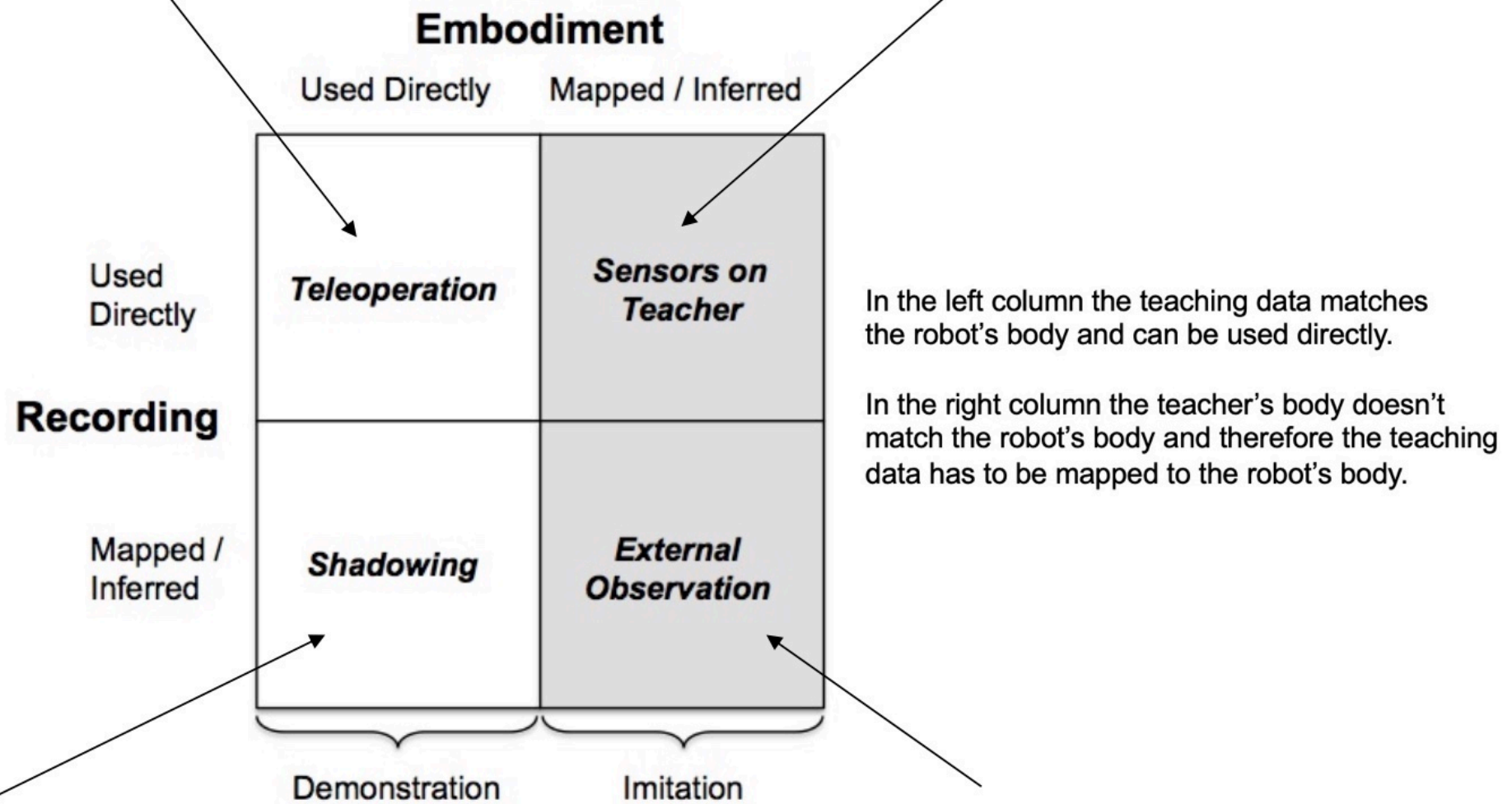
Teleoperation, Shadowing, Sensors on Teacher, and External Observation

Learning from Demonstration (LfD)

The robot has direct access to the teacher's action data

Example: human directly controls mobile robot and robot records the control data

Example: human uses gestures to teach a **mobile** robot (indirectly) and human gesture data is recorded for analysis by robot

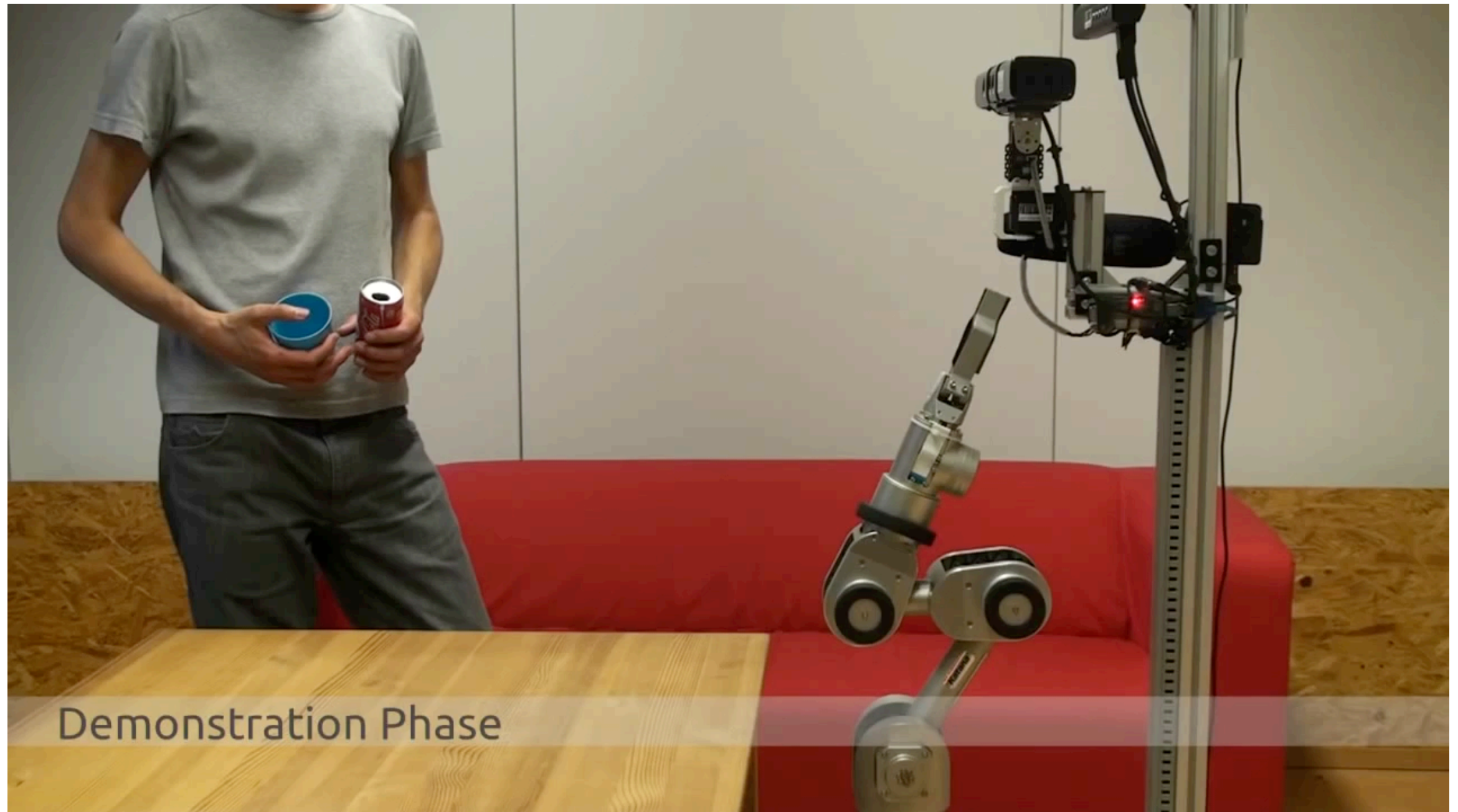


Example: human uses gestures to teach a dexterous **humanoid** robot and robot observes human, interpreting movement

The robot observes recordings of an expert performing the task

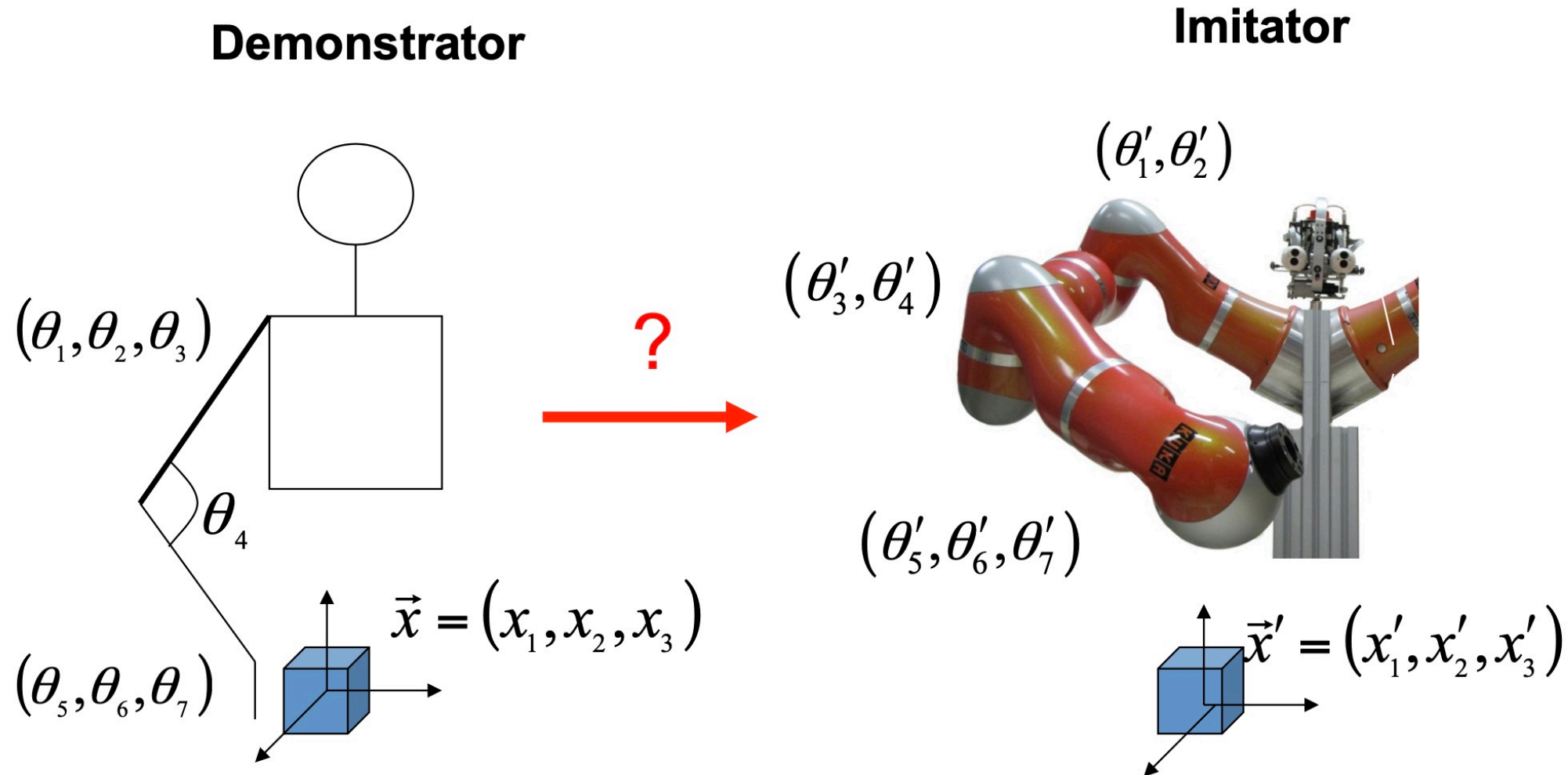
The robot doesn't have direct access to the teacher's action data

Learning from Demonstration (LfD)



Learning from Demonstration (LfD)

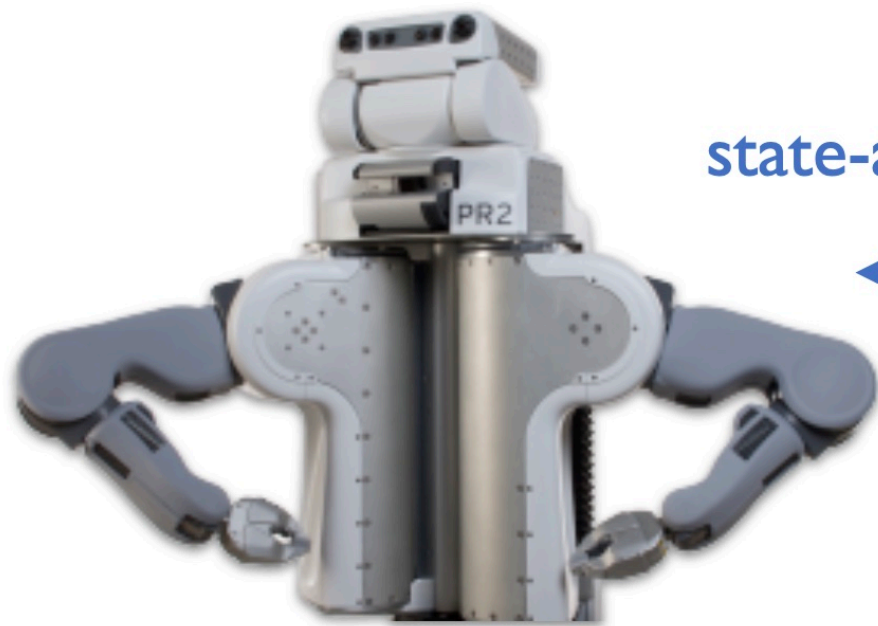
Correspondence Problem



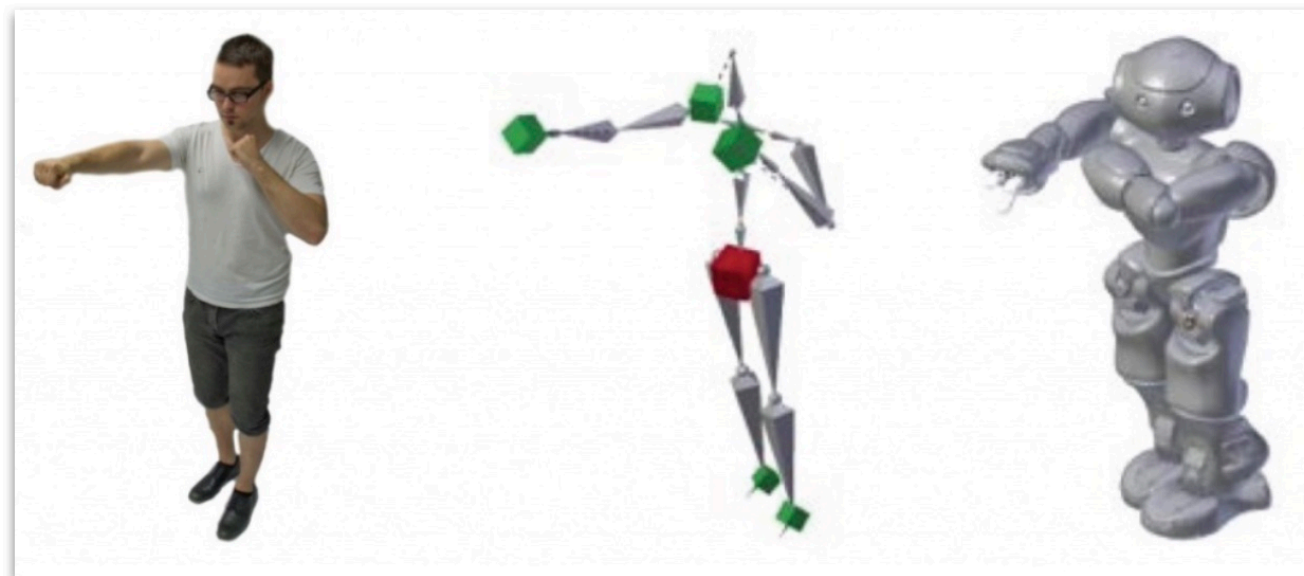
Establish a correspondence across degrees of freedom when feasible.

Learning from Demonstration (LfD)

Correspondence Problem

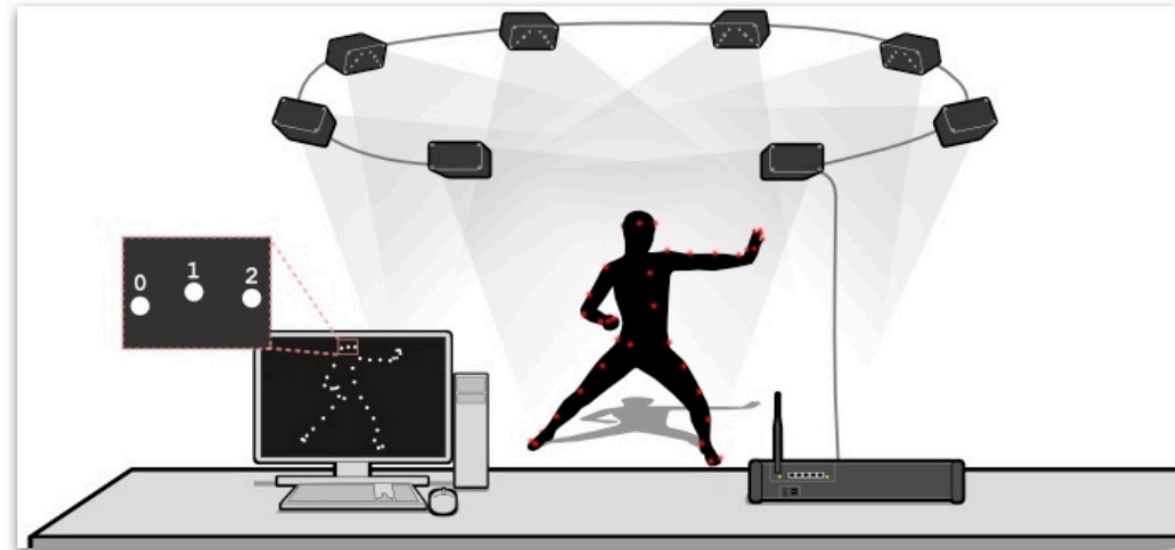
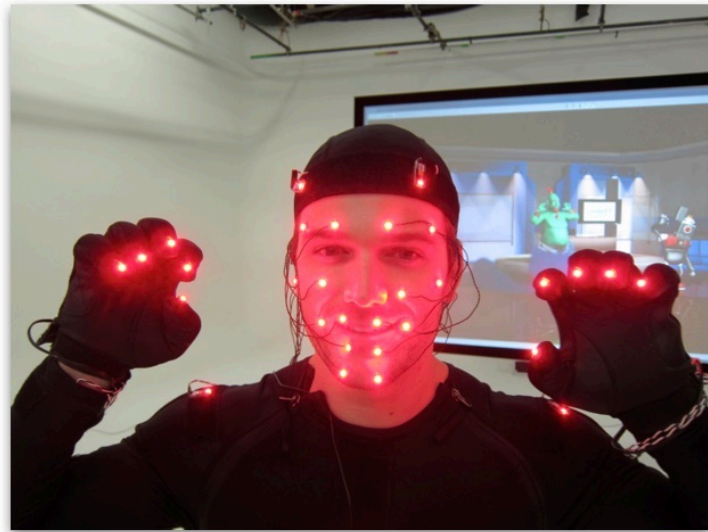


state-action mapping?

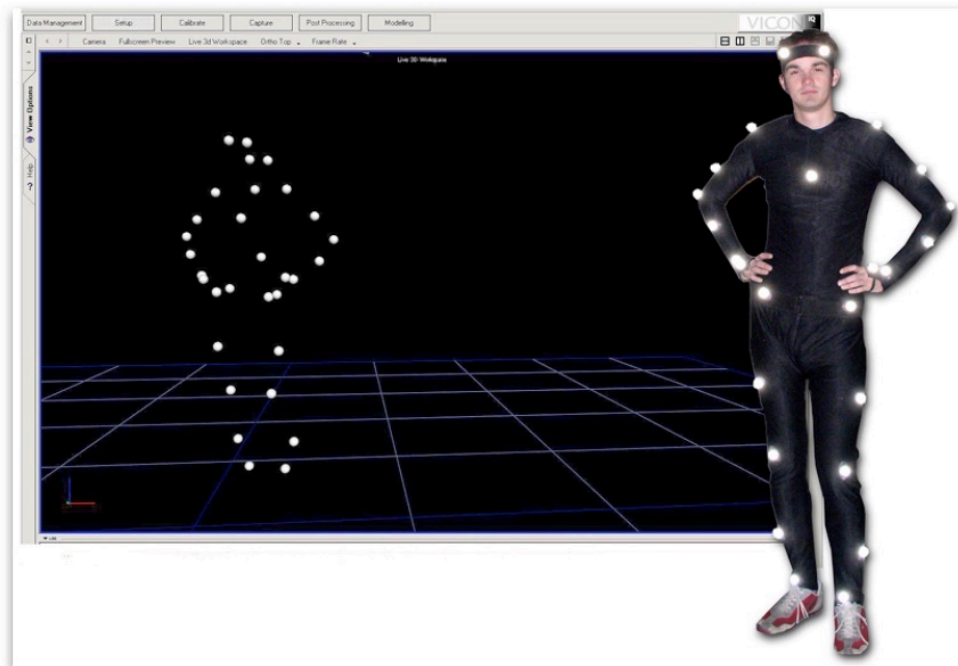


Learning from Demonstration (LfD)

Sensing: Motion capture



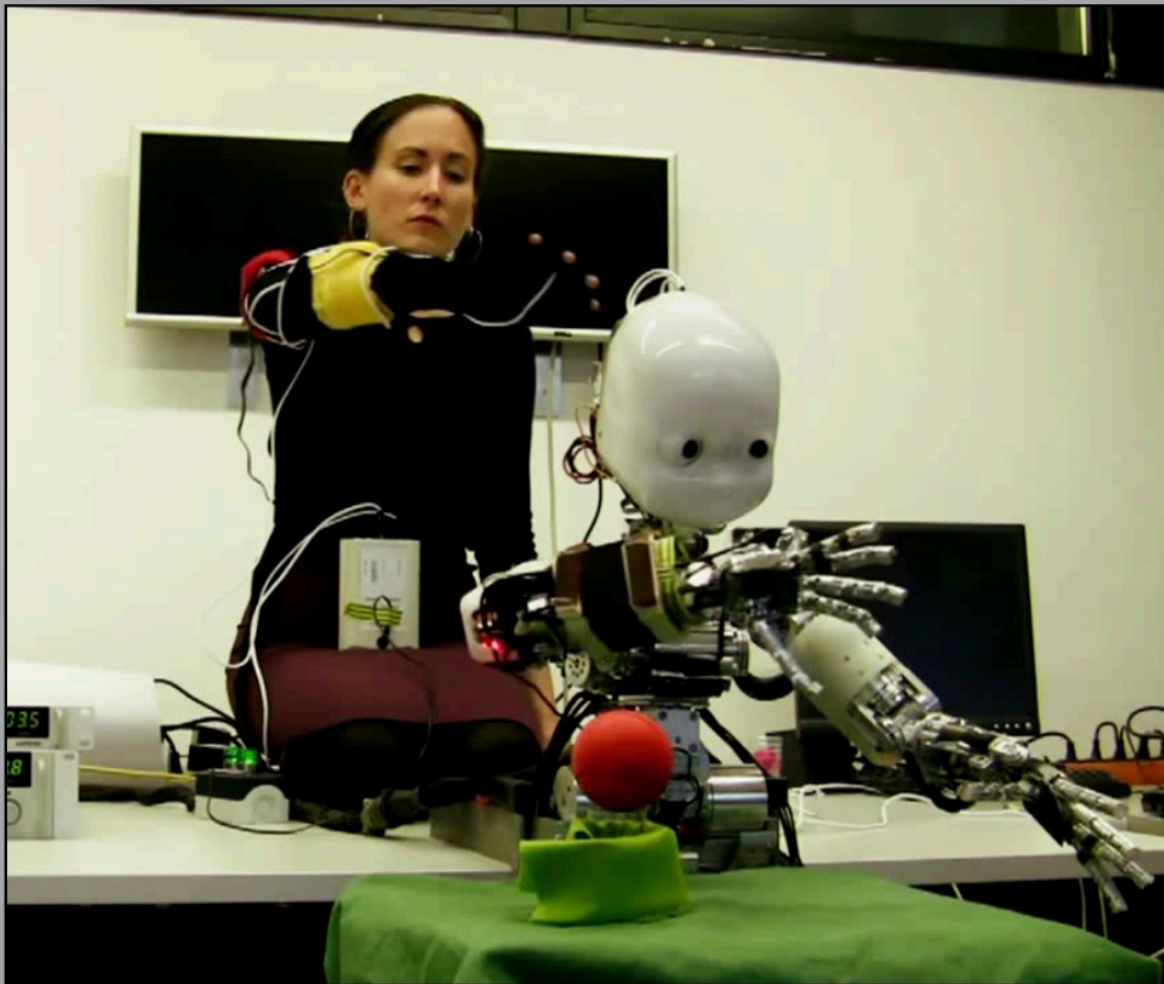
Phasespace



Vicon

Learning from Demonstration (LfD)

Which interface?



Motion sensors:

Pros:

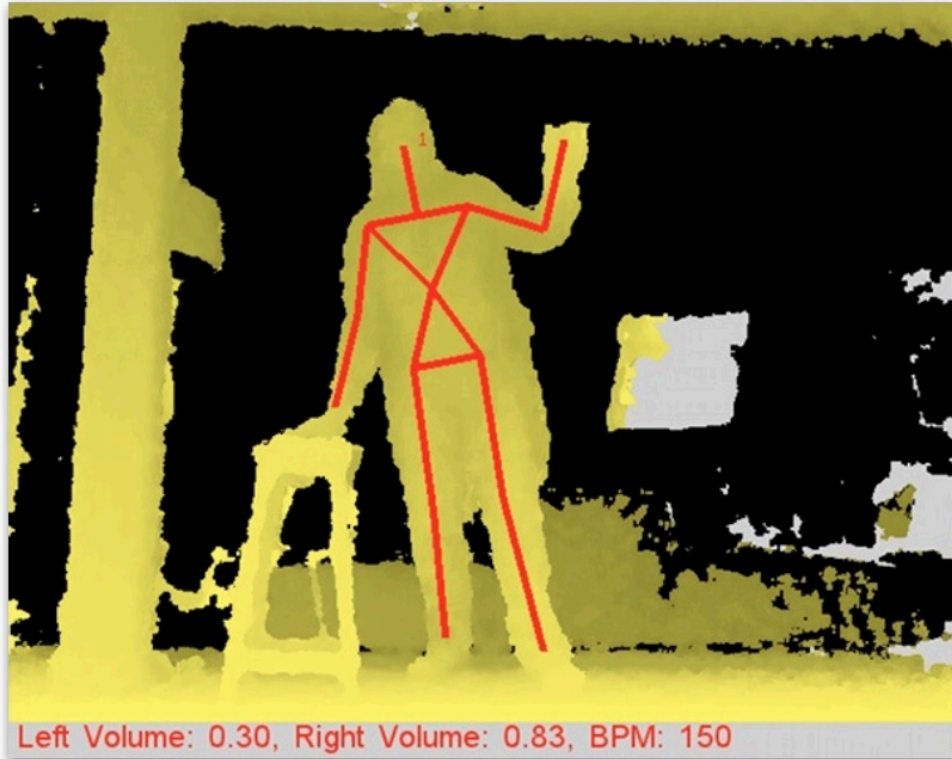
- Real-time kinematic information
- Solve correspondence problem

Cons:

- Require to wear the system
- No haptic information

Learning from Demonstration (LfD)

Sensing: RGB(D) cameras, depth sensors

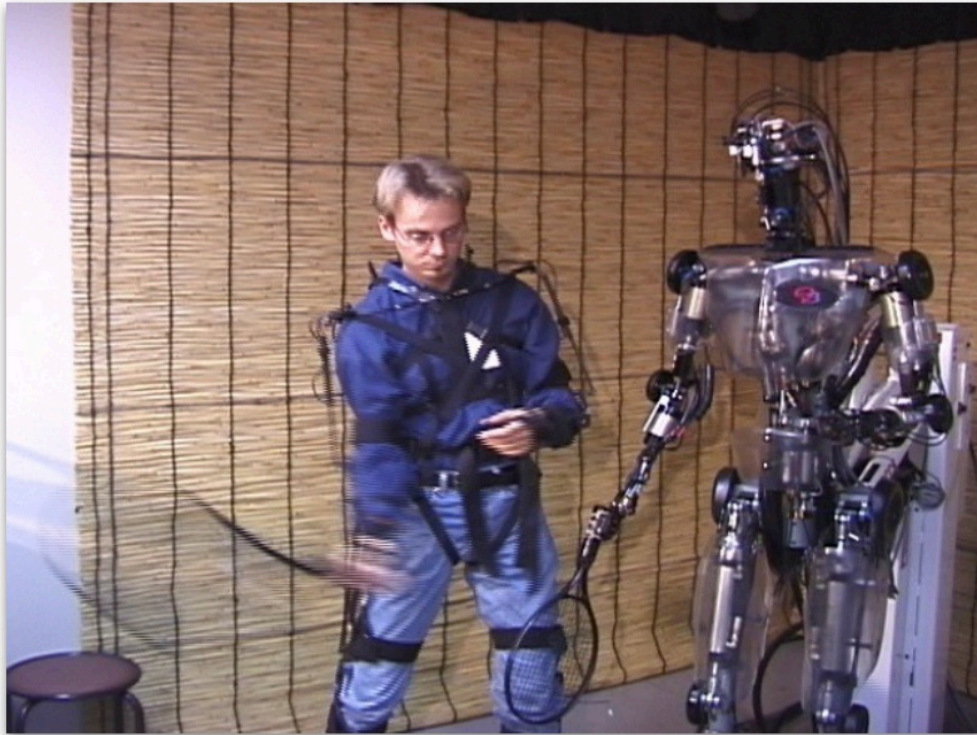


- Standard RGB cameras
- Stereo: Bumblebee
- **RGB-D: Microsoft Kinect**
- Time of flight: Swiss Ranger
- LIDAR: SICK



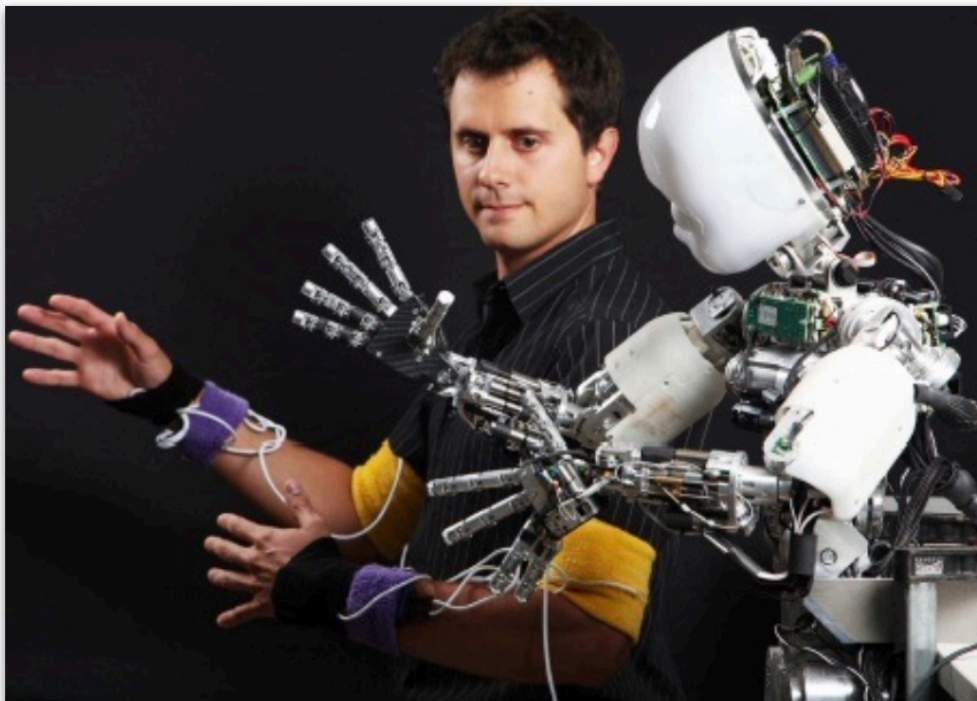
Learning from Demonstration (LfD)

Sensing: Wearable sensors



SARCOS Sensuit:

Record 35-DOF poses
at 100 Hz

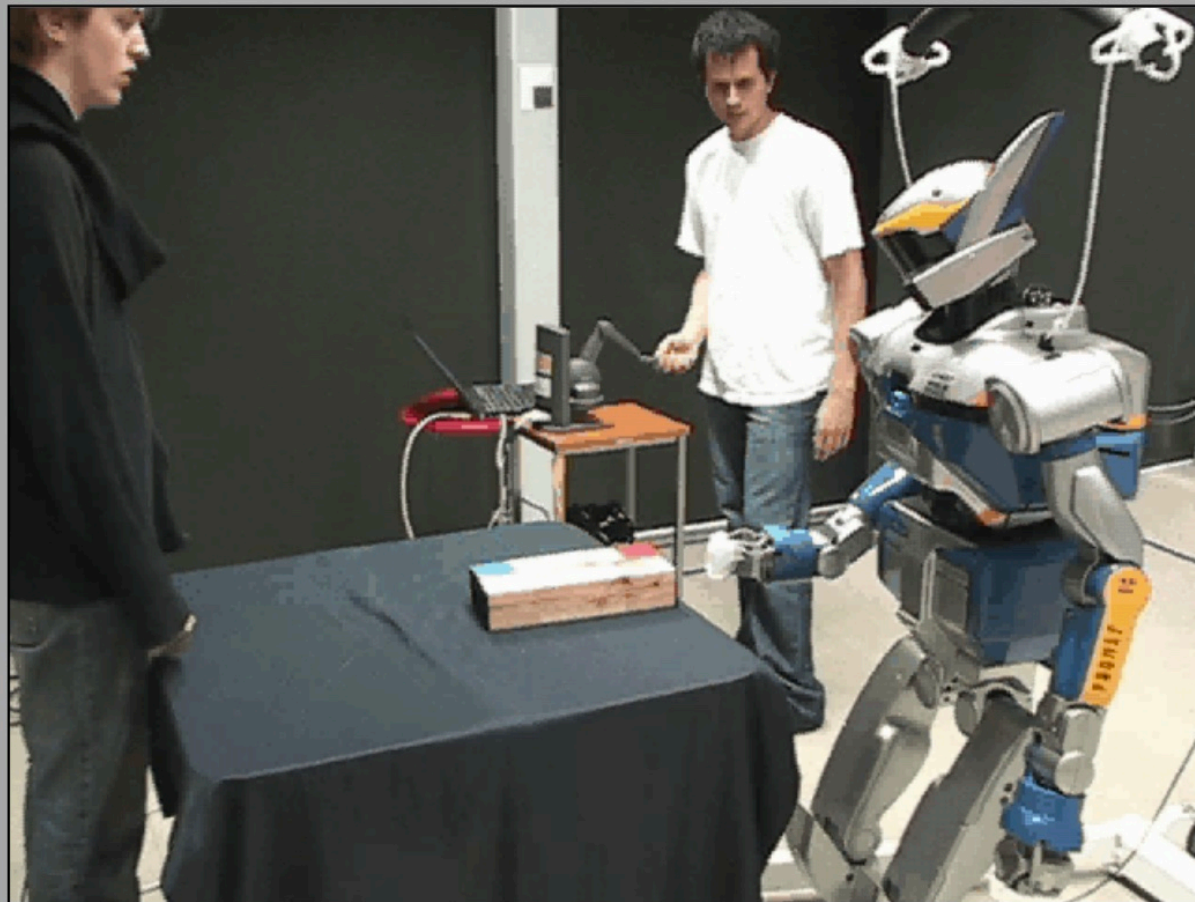


Other wearables:

- Accelerometers
- Pressure sensors
- First-person video

Learning from Demonstration (LfD)

Which interface?



Haptic devices:

Pros:

- Solve correspondence problem
- Transmit kinematic & haptic information

Cons:

- Requires training
- User far from task location

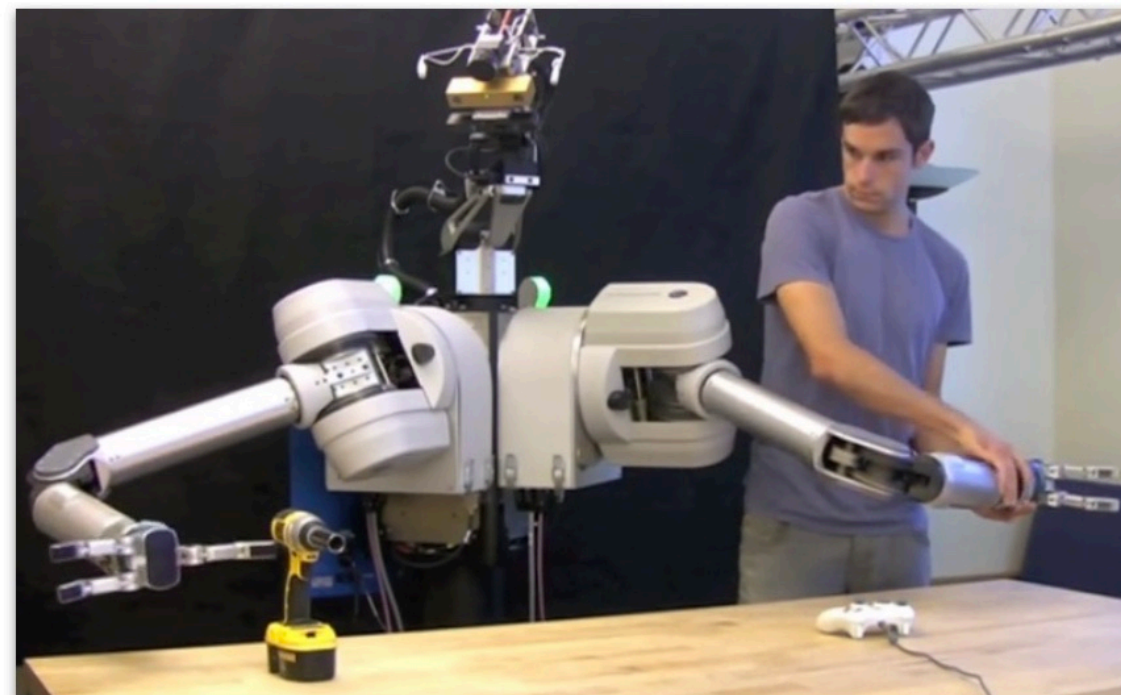
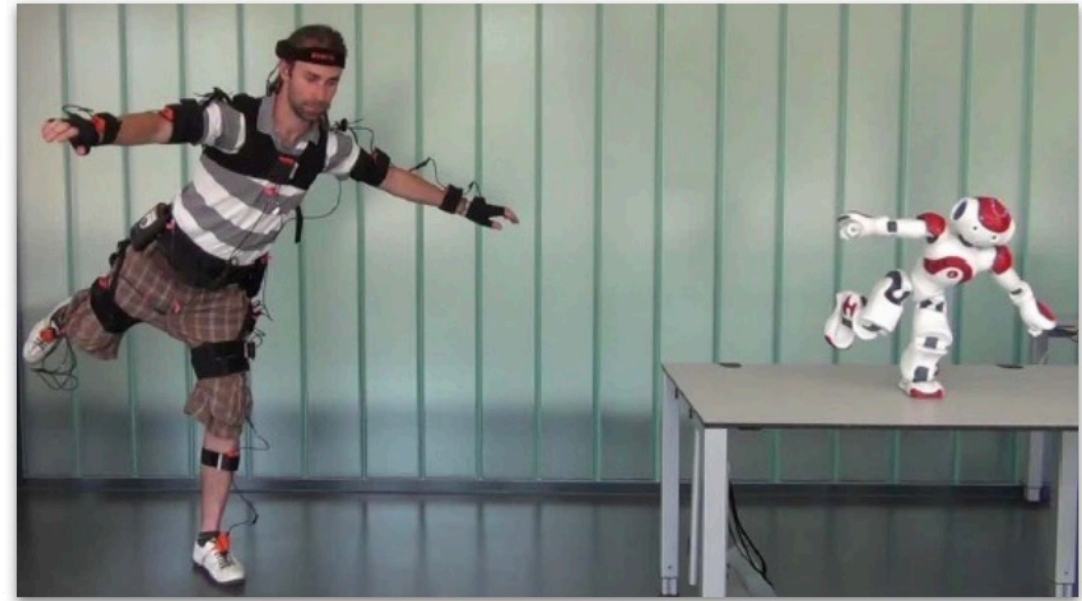
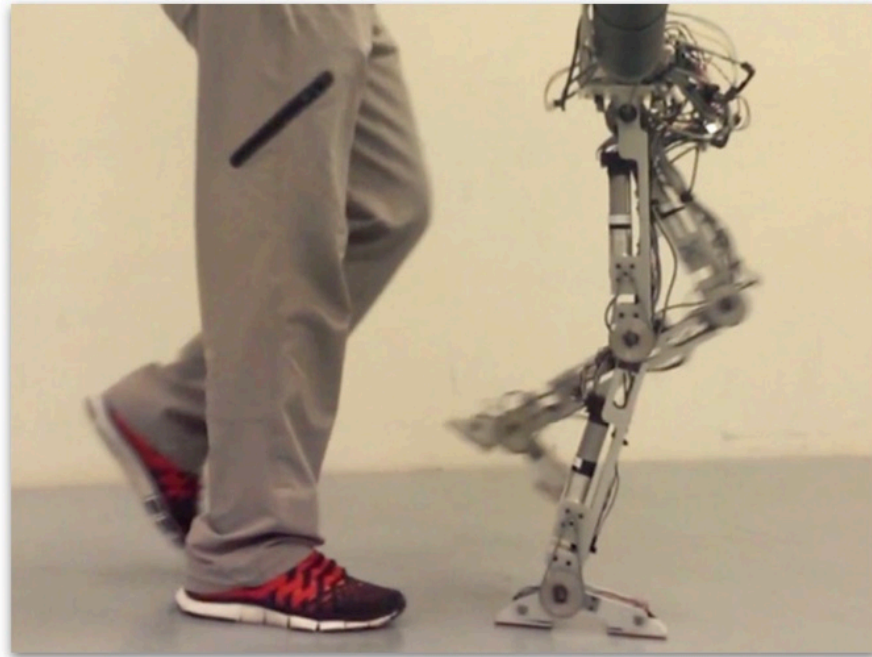
Learning from Demonstration (LfD)

Learning by doing: Teleoperation

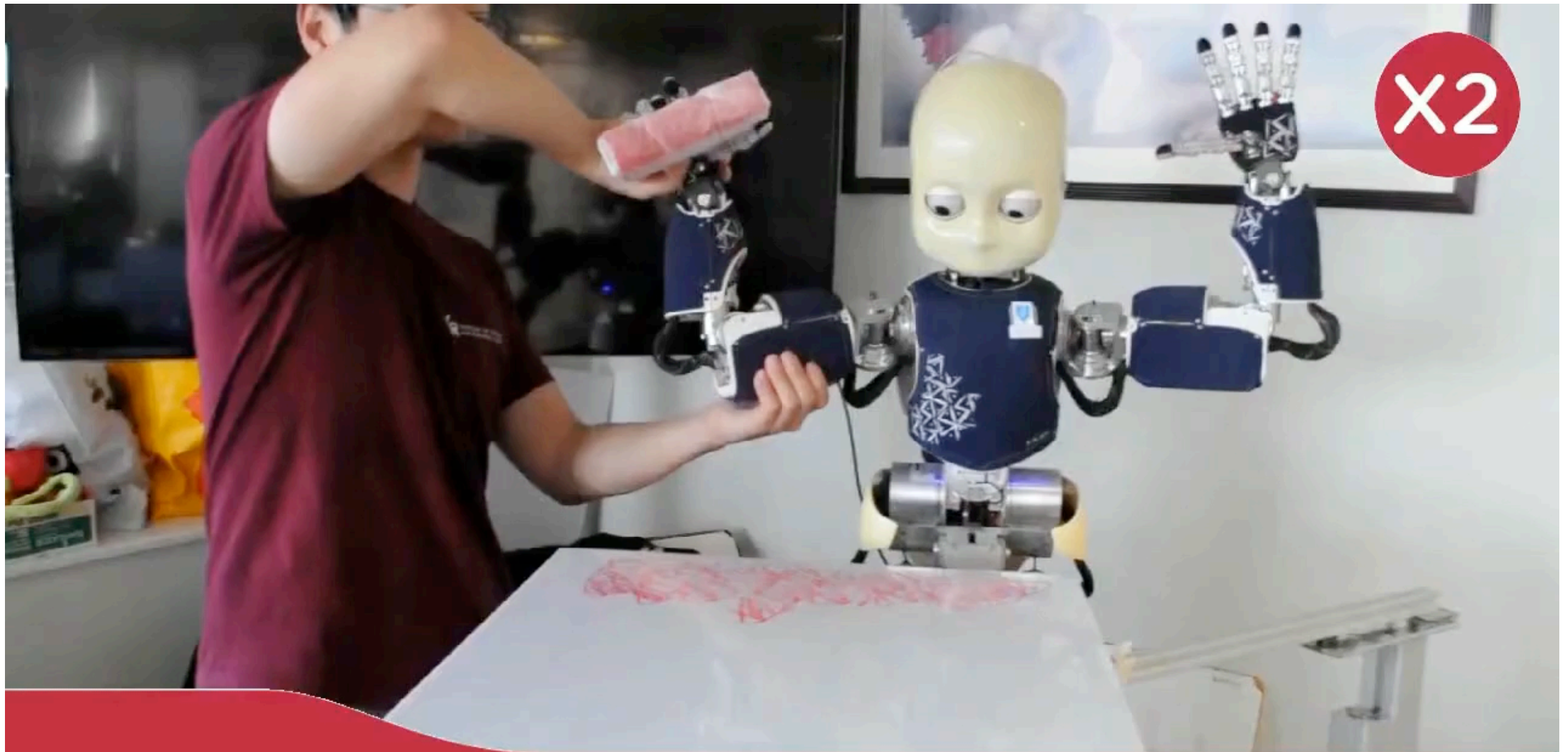


Learning from Demonstration (LfD)

Learning by watching: Shadowing



Learning from Demonstration (LfD)



In this work we taught the iCub robot how to clean a table

Summary

- Today we discussed

- 1- Learning from demonstration

- Next week

- 1- We will discuss the cognitive architectures